

D-ary Codes

Problem ID: problem-36-23810308

You have discovered a repository of classic texts from the planet Earth, which you believe to hold great archeological (and financial) value. Unfortunately, as all forms of digital storage were outlawed after the AI apocalypse, your storage capacity is very limited. Moreover, while the source text is encoded using binary 0s and 1s, your collection of black-market storage devices use *multi-valued logic*. For example, your ternary computer drive (base-3) uses ternary digits or ‘trits’, where each value can be 0, 1 or 2.

Can you find the optimal variable-length code for compressing a given text using d distinct symbols?

Input

The first line contains two integers n and d , where n is the number of lines in the text, and d is the number of symbols used by your storage drive. Each of the next n lines are strings of Unicode characters, where each line contains a maximum of 80 characters, inclusive of the newline character `\n`.

Problem bounds are $1 \leq n \leq 10000$, and $3 \leq d \leq 10$.

Important: In Windows, line endings are represented by two characters, `\r\n`. To avoid any operating system-specific issues, you should ignore any carriage return characters `\r`.

Output

Output the length of the encoded text using an optimal variable-length code for an alphabet of size d .

Sample Input 1

```
1 3
Fair is foul, and foul is fair.
```

Sample Output 1

```
73
```

Sample Input 2

```
2 3
Remember, that I am thy creature: I ought to be thy Adam; but I am rather the
fallen angel, whom thou drivest from joy for no misdeed.
```

Sample Output 2

```
374
```